AI ASSIGNMENT 1 – FOOTBALL ANALYSIS (

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1. Number of Matches

```
df.shape
```

Explanation: This returns number of rows and columns. Rows represent matches.

Interpretation: The dataset is large, meaning analysis is reliable.

2. Date Range

```
df['date'] = pd.to_datetime(df['date'])
df['date'].min(), df['date'].max()
```

Interpretation: Matches range from 1872 to 2024, covering long history.

3. Unique Countries

```
len(set(df['home_team']).union(set(df['away_team'])))
```

Interpretation: Many countries participate → football is global.

4. Average Goals

```
df['total_goals'] = df['home_score'] + df['away_score']
df['total_goals'].mean()
```

Interpretation: Around 2–3 goals → football is low scoring.

5. Home vs Away Goals

```
df['home_score'].sum(), df['away_score'].sum()
```

Interpretation: Home goals are usually higher → home advantage exists.

6. Match Results

```
def match_result(row):
    if row['home_score'] > row['away_score']:
        return 'Home Win'
    elif row['home_score'] < row['away_score']:
        return 'Away Win'
    else:
        return 'Draw'

df['result'] = df.apply(match_result, axis=1)
```

Interpretation: Used to classify match outcomes.

7. Home Win Percentage

```
home_wins = df[df['result']=='Home Win'].shape[0]
(home_wins/df.shape[0])*100
```

Interpretation: Home wins are highest → confirms home advantage.

Conclusion

Football matches are mostly low scoring. Home teams perform better. Few teams dominate globally.